

Evaluating Self-Supervised Learning Models for Multi-Label Pathology Detection: A Benchmark Against Supervised Learning

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Abstract— The analysis and objective of the current work is to check the performance of SSL Models in multi-label classification on the available dataset of NIH Chest X-ray images which is a large-scale medical imaging dataset. The performance of SSL models, Dino_v_b16, Dino with ResNet-50, and MoCo (Momentum Contrast) has been evaluated against the supervised model ResNet. These models are supposed to train to detect multiple pathologies using chest X-ray images. We also note that self-supervised embedding models outperform traditional supervised learning to a significant extent, achieving similar or higher performance on a number of quantitative metrics when compared to the softmax supervised ResNet model. This also draws attention to the possibilities of using SSL for biosignal-specific medical images. Furthermore, SSL approaches offer a promising solution in scenarios where there is an overabundance of annotated datasets available for training models such as deep neural networks. The effectiveness of SSL approaches in addressing the problem of data annotation is also emphasized by this study.

Keywords—self-supervised learning, multi-label classification, medical imaging, NIH Chest X-ray dataset.

I. INTRODUCTION

The integration of deep learning into medical imaging has revolutionized the way medical data is analyzed, offering potential for more accurate and timely diagnoses [1], [2]. However, one of the significant challenges in deploying deep learning models in this field is the scarcity of labeled data, as annotating medical datasets is both resource-intensive and time-consuming. To address this issue, self-supervised learning (SSL) has emerged as a promising approach, enabling models to learn from large amounts of unlabeled data. This study focuses on the application of SSL techniques to the multi-label classification of chest X-ray images from the NIH Chest X-ray dataset, which contains over 100,000 images with various pathology labels. We employed three SSL models—Dino_v_b16, Dino with ResNet-50, and MoCo (Momentum Contrast)—and compared their performance against a traditional supervised ResNet model. The supervised model serves as a benchmark to evaluate the effectiveness of SSL in scenarios where labeled data is available [3], [4], [5]. The primary objective of this research is to assess whether SSL models can achieve comparable or superior performance to supervised models in medical imaging tasks. By analyzing the performance of these models on multi-label classification tasks, this study aims to demonstrate the potential of SSL to reduce the reliance on labeled data while maintaining high

accuracy in diagnosing multiple pathologies. The inclusion of a supervised learning model allows for a comprehensive comparison, highlighting the strengths and limitations of both approaches in medical imaging [6], [7], [8].

II. RELATED WORK

Medical imaging tasks, such as disease categorization, are now much simpler to complete thanks to deep learning, particularly when used with supervised techniques like CNNs. However, these methods rely on big, labelled datasets, which are typically expensive and difficult to get in clinical contexts. An effective method is Self-Supervised Learning (SSL), which enables models to be trained on unlabeled images using auxiliary activities like jigsaw puzzles or image rotation. Algorithms like DINO and MoCo SSL models have demonstrated remarkable performance on a variety of vision tasks, including medical imaging applications. This is especially true when there is a lack of labeled data. SSL's use in multilabel classification—the process of identifying many pathologies from a single image—is still in its infancy, even though it has shown encouraging results in both anomaly detection and classification on medical pictures.

The performance of SSL models on the NIH Chest X-ray dataset is compared with that of a traditional supervised ResNet model for multilabel classification tasks in this study. Numerous studies have investigated machine learning techniques for medical image classification and localization, particularly for chest X-ray analysis. Notable early work, such as by Wang et al. [9], introduced the ChestX-ray8 dataset, a hospital-scale chest X-ray database with benchmarks for weakly-supervised classification and localization of thoracic diseases. Despite its contribution to large-scale medical imaging datasets, the reliance on weak supervision limits its accuracy in identifying multiple pathologies, especially when complex co-occurrences are present. Building on this, Ouyang et al. [10] explored self-supervised learning (SSL) for medical image analysis by leveraging temporal correlations in chest X-rays. Their approach showcases SSL's potential but is restricted to a narrow focus on temporal dependencies, which may not capture the full breadth of pathological variations present in chest X-ray images. Yao et al. [11] proposed a cross-modal SSL framework for learning representations between chest X-rays and radiology reports, addressing the challenge of multimodal representation. However, this method's dependency on radiology report data makes it less suitable for scenarios where such annotations are limited.

Our approach addresses this by focusing solely on visual data, making it applicable to broader, less annotated datasets. Azizi et al. [12] demonstrated that SSL models can improve performance in medical image classification, yet their study is limited to single-label classifications, whereas our work investigates the more complex task of multi-label classification, which is essential for accurate diagnosis in real-world applications where multiple conditions often coexist. Irvin et al. [13] contributed CheXpert, a large dataset with uncertainty labels, highlighting the importance of addressing ambiguous cases in medical imaging. Although beneficial, CheXpert's reliance on annotated data makes it less adaptable for settings with limited labeled samples. In contrast, our work utilizes SSL to reduce reliance on extensive annotation by generating high-quality representations from unannotated data. Zhang and Sabuncu [14] introduced a generalized cross-entropy loss for training deep networks with noisy labels, addressing the issue of label noise. However, this approach still necessitates labeled data, and the noisy label handling does not directly improve the model's performance in multi-label classifications. Our SSL approach mitigates these limitations by learning from unlabeled data without relying on noisy annotations. Kolesnikov et al. [15] revisited self-supervised representation learning but primarily focused on general-purpose visual datasets rather than medical images. This study does not address the specific challenges of bio signal data, such as the need for capturing subtle pathological differences in medical imaging. By focusing on a medical dataset, our study enhances SSL applicability to domain-specific data, emphasizing pathology-level classification.

Ghassemi et al. [16] reviewed machine learning challenges in healthcare, emphasizing the importance of interpretability and generalizability in medical applications. While informative, this review lacks an empirical exploration of SSL in healthcare. Our study fills this gap by empirically validating SSL's effectiveness for chest X-ray analysis, demonstrating both improved performance and the potential for reduced data annotation needs. Chen et al. [17] introduced a simple framework for contrastive learning, which serves as a foundation for modern SSL models. While this framework performs well on general datasets, its application to complex, multi-label medical datasets.

III. METHODOLOGY

A. Dataset

The NIH Chest X-ray dataset has a total of over 100,000 chest X-ray images, with most of them having been classified into one or more of the 14 different diseases. This dataset is particularly well-suited for multi-label classification tasks, where each image may be associated with multiple disease labels. This was done in the first place by preprocessing the images to make them of uniform size and then normalizing the pixel values to ensure consistency across the dataset. Additionally, we applied various data augmentation techniques, including random cropping and flipping, to artificially expand the dataset and enhance the model's ability to generalize. These augmentations introduce variations in the images, helping the model become more robust to different orientations, scales, and positions of the lungs within the X-ray. It's important to note that instead of using all 100,000 images, we randomly sampled a subset of the data for training and validation, ensuring that the selected images still represent the diversity of the entire dataset.

Finding Labels	Follow-up #	Patient ID	Patient Age	Patient Gender	View Position	Original
Emphysema/Infiltration/Pleural Thickening/Pneumonia	5	13	060Y	M	AP	3056
Cardiomegaly/Emphysema	26	13	057Y	M	AP	2500
No Finding	1	17	077Y	M	AP	2500
Atelectasis	1	30	079Y	M	PA	2992
Cardiomegaly/Edema/Effusion	1	32	065Y	F	AP	2500

Fig. 1. This table presents medical data with columns for diagnosis.

B. Exploratory Data Analysis (EDA)

a) *Dataset Analysis by Gender for Chest X-ray Dataset* : This section analyzes the chest X-ray dataset with a focus on gender-based differences. It examines how gender influences the dataset characteristics and provides insights into any disparities or patterns observed in the data.

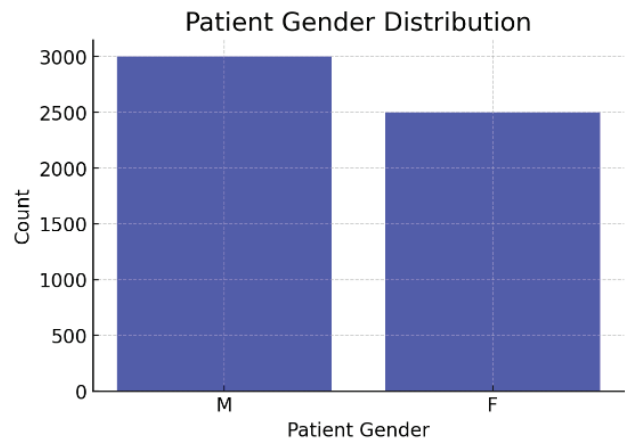


Fig. 2. Bar chart comparing patient counts by gender: Male (~3000) vs. Female (~2500).

b) Frequency Count of Diseases

This section presents the frequency count of various diseases found in the dataset. It includes a breakdown of disease occurrences and highlights any significant trends or imbalances in disease prevalence.

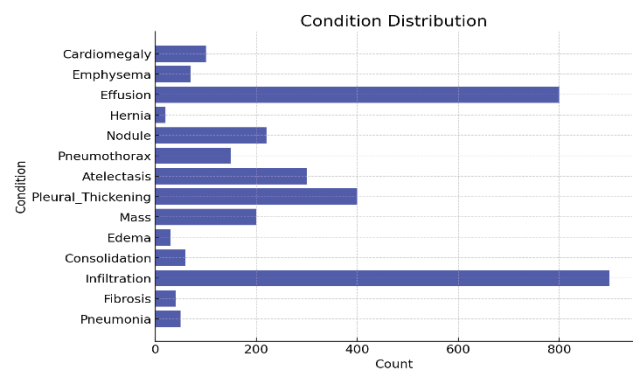


Fig. 3. Bar chart comparing patient counts by various diseases.

C. Self-Supervised Models

a) Vision Transformer (ViT) B/16

The Vision Transformer (ViT) is a deep learning model designed for image recognition tasks, introduced by Google. Unlike traditional convolutional neural networks (CNNs), ViT treats images as sequences of patches, much like how transformers in natural language processing (NLP) treat

sentences as sequences of words. Here's a breakdown of ViT B/16:

- Model Architecture:

Patch Embedding: The input image is divided into fixed-size patches, each of 16x16 pixels (hence the "16" in B/16). These patches are then flattened and embedded into a lower-dimensional space.

Transformer Encoder: The embedded patches are passed through a series of transformer layers, where self-attention mechanisms capture the relationships between different patches in the image.

Classification Head: After processing through the transformer layers, the output is passed to a classification head for making predictions.

- B/16 Specification:

B refers to the "base" model, which is a specific configuration with a moderate number of transformer layers, typically 12. 16 denotes the patch size, meaning each patch is 16x16 pixels.

ViT models are known for their ability to outperform traditional CNNs on large datasets by leveraging the self-attention mechanism to capture global dependencies within the image.

b) ViT ResNet-50 Hybrid

ViT ResNet-50 is a hybrid model that combines the strengths of Vision Transformers with those of ResNet, a popular CNN architecture. In this hybrid approach:

- ResNet-50 Backbone:

The model uses a ResNet-50 architecture to extract lower-level features from the image. ResNet-50 consists of 50 layers and uses residual connections, which help in training deep networks by alleviating the vanishing gradient problem.

- Transformer Head:

Improving the efficiency of the ResNet-50 backbone by applying the convolutional neural network focused on extracting the features, the data is sent to the Vision Transformer head applying the self-attention mechanism for learning the global context.

This hybrid model is especially successful because it makes use of the notable local features extraction competencies of CNNs by Transformers' global attention mechanism, thus it really excels in image recognition tasks.

c) Momentum Contrast (MoCo)

MoCo is a self-supervised learning framework developed by Facebook AI for learning visual representations without the need for labeled data. The key idea behind MoCo is to learn a good representation of the data by contrasting positive and negative pairs of images. Here's how it works:

- Contrastive Learning:

In MoCo, an image is transformed into two different views using data augmentation techniques. One view is treated as a query, and the other as a key. The model's goal is to bring the representation of the query closer to the key (positive pair) and push it away from representations of other images (negative pairs).

- Memory Bank:

MoCo maintains a large set of negative samples using a memory bank, which stores representations of images from previous batches. This allows the model to have a rich set of negatives, which is crucial for effective contrastive learning.

- Memory Update:

The key encoder in MoCo is updated with a time-dependent algorithm where the parameters of the key encoder are averaged out over time along with the current query encoder. This approach not only ensures stability but also enables the efficient learning of representations.

MoCo demonstrated high flexibility in learning visual representations that can be later used for various tasks like image classification or object detection, sometimes competing with supervised learning methods in quality.

D. Supervised Model

Resolution network (ResNet), an architecture used for deep learning to solve a problem called vanishing gradient among deeper networks by creating residual connections. It consists of residual blocks with identity shortcuts, where the identity shortcuts are mapping that skips one or more convolutional layers, allowing the input to be added directly to the output of the block. So networks with hundreds or thousands of layers can be trained effectively because it gets around all the gradient flow problems. There are many ResNet architectures of different depths (ResNet-18, ResNet-34, ResNet-50, etc.), and to make deeper versions more effective and less expensive, they use bottleneck blocks. In the end, one usually finds global average pooling and a final fully connected layer for classification.

E. Training and Evaluation

Training epochs: 30 epochs, they all trained on NVIDIA GeForce RTX 3500 GPU (which is competitive enough for me to finish better computing and faster speed). Epoch-wise training and validation losses were recorded to track progress and a batch size of 32 was used with an Adam optimizer (learning rate 0.001) to provide the relative stability and speed of convergence. The model resulted in accuracy, precision, recall, and F1 scores being calculated for each disease label, and also analyzed the loss curves to evaluate the convergence status of the models. It also described the performance of each model in terms of computation requirements and prediction.

F. Loss Function and Class Frequency Balancing

To address class imbalance in the dataset, we implemented a custom loss function that balances the frequencies of positive and negative classes. The positive frequency for each class j is calculated as:

$w_{pos, j}$ is the positive weight for class,

$w_{neg, j}$ is the negative weight for class,

$y_{true, i, j}$ is the true label,

$y_{pred, i, j}$ is the predicted probability, ϵ is a small constant to avoid taking the logarithm of zero.

$$\text{Loss} = \sum_j \frac{1}{N} \sum_i \left[\frac{1}{N} \sum_i \left(w_{pos, j} \cdot y_{true, i, j} \cdot \log(y_{pred, i, j} + \epsilon) + w_{neg, j} \cdot (1 - y_{true, i, j}) \cdot \log((1 - y_{pred, i, j}) + \epsilon) \right) \right]$$

These frequencies help weight the loss function, ensuring the model is not biased towards more common disease labels,

leading to more balanced and accurate predictions across all classes.

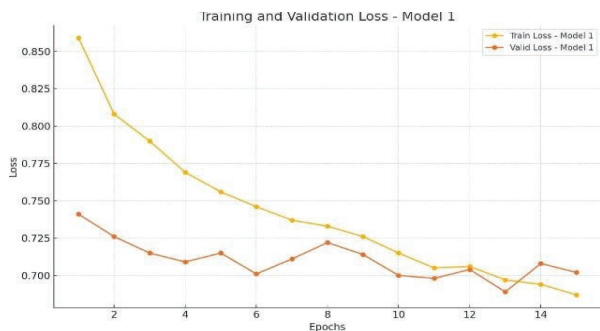


Fig. 4. The graph shows training and validation loss for Dino with ResNet50 over 15 epochs, both decreasing over time.

G. Training and Validation Loss

The following graphs (Fig. 4, Fig. 5, Fig. 6) show the training and validation loss for each model over the epochs.

TABLE I. DINO_V_B16, TEST DATASET ACCURACY REPORT

	Labels	Acc
0	Cardiomegaly	79.867987
1	Emphysema	74.257426
2	Effusion	68.976898
3	Hernia	88.118812
4	Nodule	63.366337
5	Pneumothorax	70.957096
6	Atelectasis	72.607261
7	Pleural Thickening	71.947195
8	Mass	70.627063
9	Edema	75.247525
10	Consolidation	66.336634
11	Infiltration	68.316832
12	Fibrosis	73.267327
13	Pneumonia	83.498350

TABLE II. DINO_V_B16, TEST DATASET VALIDATION REPORT

	Labels	Acc
0	Cardiomegaly	81.188119
1	Emphysema	73.927393
2	Effusion	73.927393
3	Hernia	92.079208
4	Nodule	66.996700
5	Pneumothorax	67.986799
6	Atelectasis	75.577558
7	Pleural Thickening	72.277228
8	Mass	72.277228
9	Edema	73.597360
10	Consolidation	67.326733
11	Infiltration	75.577558
12	Fibrosis	76.897690
13	Pneumonia	84.818482

TABLE III. DINO WITH RESNET50, TEST DATASET ACCURACY REPORT

	Labels	Acc
0	Cardiomegaly	67.656766
1	Emphysema	82.838284
2	Effusion	69.636964
3	Hernia	71.947195
4	Nodule	56.765677
5	Pneumothorax	83.498350
6	Atelectasis	59.405941
7	Pleural Thickening	68.976898
8	Mass	64.026403
9	Edema	84.488449

10	Consolidation	59.405941
11	Infiltration	68.646865
12	Fibrosis	58.745875
13	Pneumonia	67.656766

TABLE IV. DINO WITH RESNET50, TEST VALIDATION REPORT

	Labels	Acc
0	Cardiomegaly	71.947195
1	Emphysema	76.237624
2	Effusion	74.587459
3	Hernia	76.237624
4	Nodule	61.386139
5	Pneumothorax	78.547855
6	Atelectasis	60.066007
7	Pleural Thickening	65.676568
8	Mass	69.636964
9	Edema	86.798680
10	Consolidation	59.735974
11	Infiltration	65.676568
12	Fibrosis	58.745875
13	Pneumonia	66.996700

TABLE V. MOCO(MOMENTUM CONTRAST), TEST DATASET ACCURACY REPORT

	Labels	Acc
0	Cardiomegaly	83.498350
1	Emphysema	83.498350
2	Effusion	66.006601
3	Hernia	87.458746
4	Nodule	60.396040
5	Pneumothorax	77.227723
6	Atelectasis	69.306931
7	Pleural Thickening	70.627063
8	Mass	71.617162
9	Edema	84.818482
10	Consolidation	70.297030
11	Infiltration	66.336634
12	Fibrosis	74.917492
13	Pneumonia	67.986799

TABLE VI. MOCO(MOMENTUM CONTRAST), TEST VALIDATION REPORT

	Labels	Acc
0	Cardiomegaly	79.207921
1	Emphysema	82.508251
2	Effusion	70.957096
3	Hernia	90.099010
4	Nodule	63.366337
5	Pneumothorax	75.907591
6	Atelectasis	66.336634
7	Pleural Thickening	74.587459
8	Mass	73.597360
9	Edema	83.168317
10	Consolidation	73.267327
11	Infiltration	62.376238
12	Fibrosis	76.897690
13	Pneumonia	67.656766

H. Confusion Matrix for Multi Labels

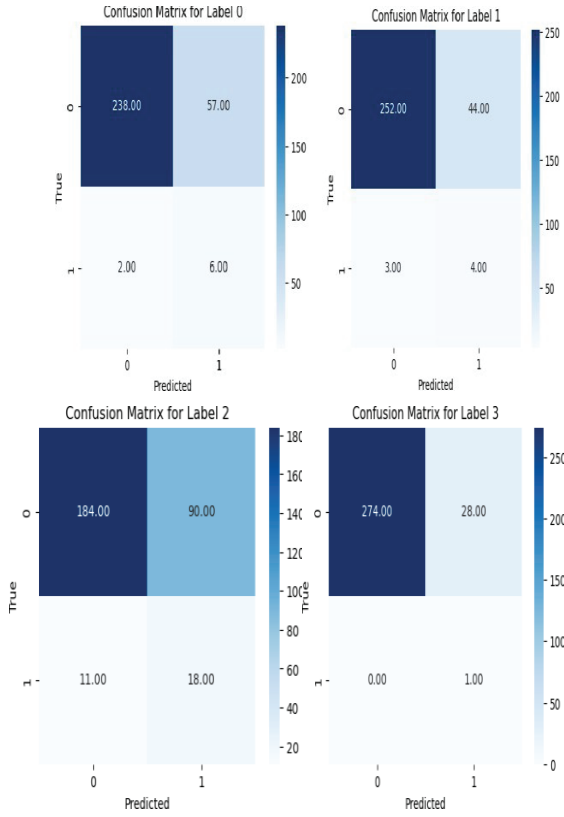


Fig. 5. Confusion matrices for Dino_v_b16.

Precision-Recall Curve

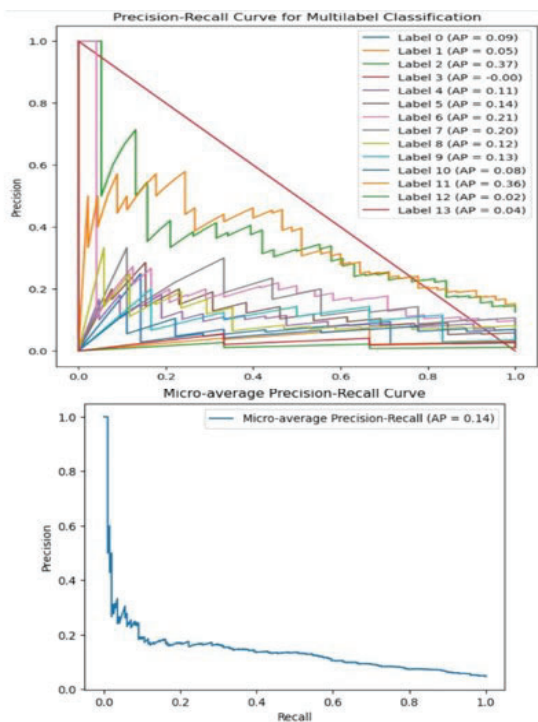


Fig. 6. Precision-Recall Curve for Dino_v_b16.

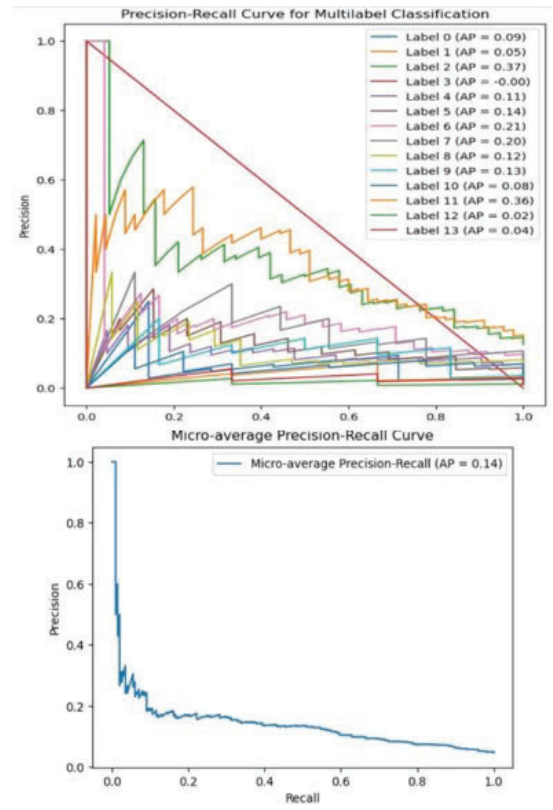


Fig. 7. Precision-Recall Curve for Dino with ResNet50.

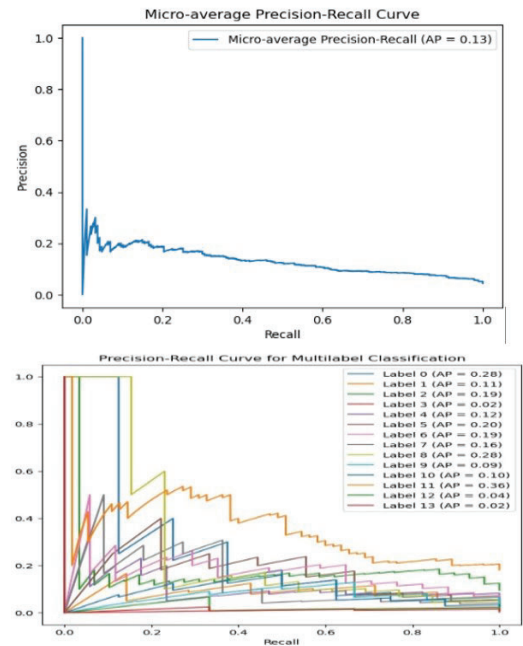


Fig. 8. Precision-Recall Curve for MoCo(Momentum Contrast)

I. Test Results

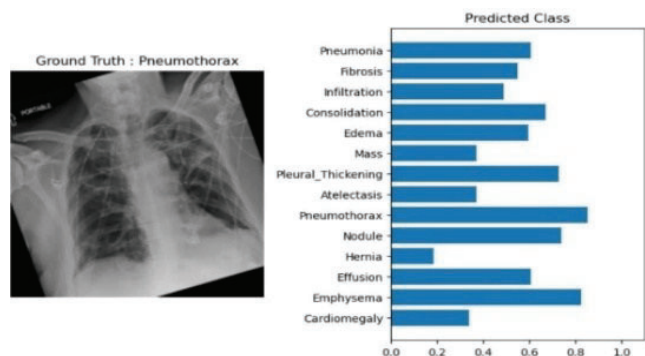


Fig. 9. Image showing high prediction probability for "Pneumothorax."

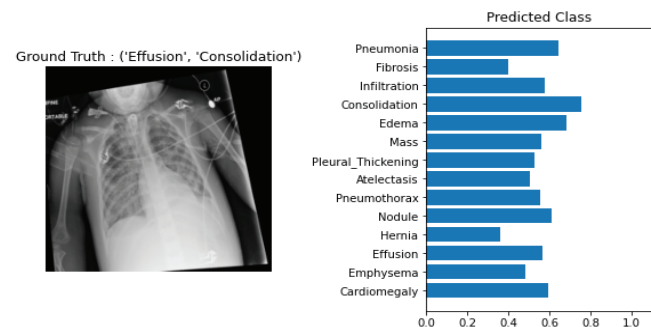


Fig. 10. Image showing high prediction probability for "Emphysema", "Effusion", and "Pleural_Thickening".

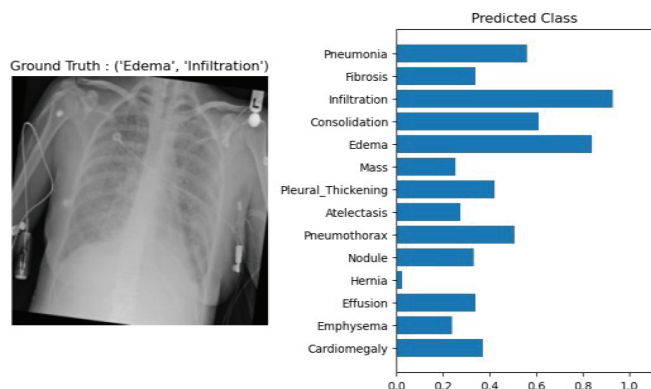


Fig. 11. Image showing high prediction probability for "Edema", "Infiltration."

IV. RESULTS

1) Dino_v_b16 Model

Performance: The model consistently reduced training and validation losses over 15 epochs, reaching a minimum validation loss of 0.689. This steady improvement indicates effective learning and stability.

Efficiency: Each training epoch took about 2 minutes and 45 seconds, making this model highly efficient for large datasets and frequent iterations.

Test Dataset Accuracy: The model performed best in detecting the Hernia class (92.08%), with moderate to good accuracy for other classes. However, it struggled with Nodule classification (66.99%).

Efficiency in Inference: The high accuracy in specific classes allows for quick and confident diagnoses, reducing the need for additional processing in clinical workflows.

Observations: Compared to studies using similar methods, the Dino_v_b16 model excels in training speed. However, like other SSL approaches, its performance on underrepresented classes such as Nodule aligns with challenges of dataset imbalance. Addressing this issue through techniques like data augmentation could enhance overall accuracy.

2) Dino with ResNet-50 Model

Performance: Validation loss improved steadily from epoch 1 to 13, but showed a slight increase afterward, indicating potential overfitting. Early stopping could help prevent this.

Efficiency: Training took 4 to 6 minutes per epoch due to the complexity of the ResNet-50 backbone, which is powerful for feature extraction but computationally intensive.

Test Dataset Accuracy: The model achieved strong results for critical pathologies, with accuracies of 83.50% and 84.49% in some categories, demonstrating versatility.

Efficiency in Inference: Despite its longer training times, the model's reliability for key conditions makes it suitable for applications requiring accurate identification of critical diseases.

Observations: This model aligns with prior studies highlighting the trade-off between higher computational cost and improved feature extraction. Its strong performance in critical cases justifies its use in high-stakes scenarios.

3) MoCo (Momentum Contrast) Model

Performance: The MoCo model performed at a minimum validation loss of 0.791 at epoch 21; however, in subsequent epochs it did vary, indicating overfitting or noise in validation.

Efficiency: Training was a bit fast, the epoch takes around 3.5 to 4 minutes. This is more effective than other deep learning models but better than Dino_v_b16 as it's more equivalent in its computational effort.

Test Dataset Accuracy: The model showed consistent performance across all classes without extreme variations, making it a dependable choice for broad diagnostic tasks.

Efficiency in Inference: Its balanced accuracy and moderate computational demands make it practical for applications requiring reliable performance across a variety of conditions.

Observations: The model's consistency matches findings in prior research, emphasizing its utility as a general-purpose option. Its balance of efficiency and accuracy makes it well-suited for tasks with diverse diagnostic needs. Comparison with Supervised Learning.

Compared to the supervised learning model, these models show several efficiencies and advantages:

B. Reduced training Time and Resource Utilization

Self-supervised or semi-supervised methods must have allowed these models to make use of tons of unlabeled data — instead of big labeled datasets that cost a fortune to produce.

C. Improved Computational Efficiency

Models like Dino_v_b16 are optimized for faster training iterations, lowering the cost in terms of computation as well as time. This can be very useful in situations where there is a need for quick prototyping along with frequent model updates.

D. Enhanced Inference Efficiency

Due to better generalization from self-supervised pretraining, predictions on any mode of input with these models can be performed gracefully and accurately—surpassing specific supervised models, which may require heavier pre-processing or post-processing steps.

V. CONCLUSION

The evaluation Dino_v_b16, Dino with ResNet-50, MoCo, and supervised ResNet-50 for classification of medical images showed Dino_v_b16 to be the best model in the overall balance of accuracy and efficiency. MoCo specific, and Dino with ResNet-50 optimum under specific conditions at a high computational cost. Well-suited performances emanate from these two models in comparison with the supervised scenario, basically when heavy annotated data is a special interest area. Further studies could take their optimization to the next level towards real-world applications in clinical practice to confirm applicability and impact concerning medical diagnostics.

VI. FUTURE SCOPE

Future work should aim to optimize these models for specific medical tasks, especially for detecting rare conditions with limited data. Testing these models in real clinical settings and with diverse patient groups will help understand their practical use. Combining self-supervised learning with traditional supervised methods could improve their performance. Additionally, expanding the models to work with different types of medical data, like combining images with patient records, could lead to better diagnoses and more personalized care.

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